

6. MASS AND BALANCE

6.1 EQUIPMENT LIST ENTRIES

System/Equipment Designation	Part No.	Mass (kg)	Arm (mm)	Massmoment (kgmm)
Fuel Microfilters with Back-up Supply System	105-83561 105-83573	6.2	3900	24180

7. SYSTEM DESCRIPTION

Installation of the fuel microfilters with back-up supply system provides additional fuel system protection while guaranteeing positive fuel pressure to each engine.

7.1 FUEL MICROFILTERS

A microfilter is installed in each engine's fuel supply system between the emergency fuel valve and the engine-mounted fuel filter. The microfilter will entrap not only contaminants but also ice crystals that may form at fuel temperatures below +4°C so that the requirement for a fuel anti-icing additive can be eliminated.

Each microfilter is fitted with a differential pressure switch that will activate the respective **FILT** warning light when a differential of 1.2 to 1.4 psi is sensed, indicating that clogging of the filter element has begun. When the pressure differential reaches 4.0 psi (filter clogged) a bypass valve opens allowing unfiltered fuel to flow downstream.

NOTE The **FILT 1/FILT 2** warning lights will also come on whenever their respective engine-mounted fuel filter becomes clogged.

7.2 BACK-UP SUPPLY SYSTEM

The back-up supply system integrates the main tank pumps into the fuel supply system in a standby role. Each main tank pump connects to a supply tank pump through fuel lines fitted with a pressure relief valve and two check valves.

Under normal conditions, the higher supply tank pump output pressure maintains the check valve in the fuel line from the main tank closed since the main tank pump pressure is limited to 0.5 kp/cm² by the pressure relief valve.

Should the supply tank pump fail, the check valve will open to allow main tank fuel to flow directly to the engine concerned. A second check valve at the supply tank pump prevents the main tank fuel from flowing into the supply tank through the inoperative pump.

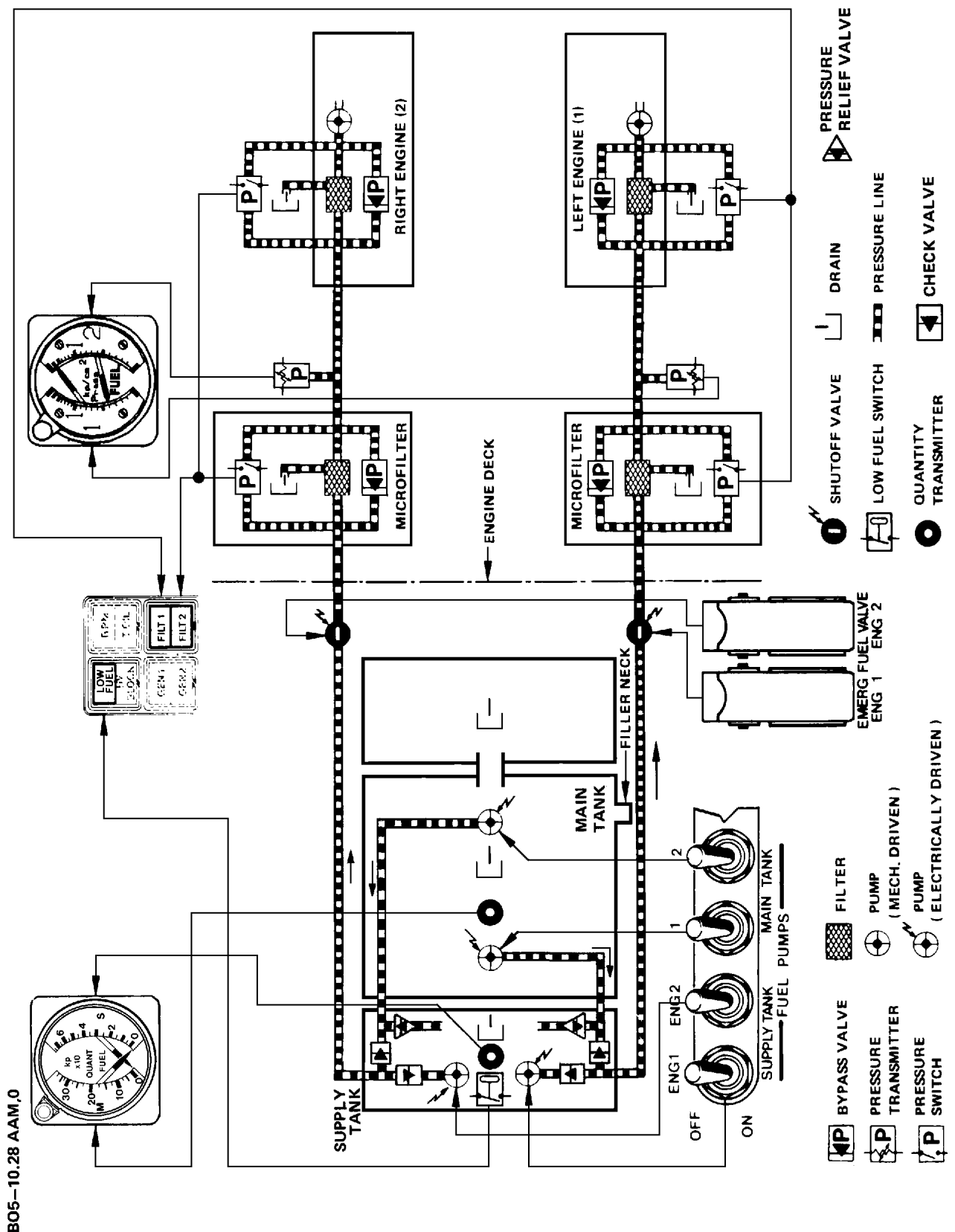


Fig. 1 Fuel Microfilters with Back-up Supply System

